

PO Notes New Persistency

Architecture Comparison & Recommendation Guide

Notes Reuse RAP | Sales SDD Own Table | ZPO_LONGTEXT | SAPScript Handler | SDM

SAP S/4HANA Procurement | EKKO / EKPO | CE2702 / CE2708 Planning | 2026-06-30

What This Document Covers

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Context: CE2702/CE2708 sync-up meeting (2026-06-25) identified 3 persistency options for PO notes migration. Sales team completed their migration in CE2608 as the reference implementation. The critical open question — can application-specific own table work with Notes Reuse RAP — is pending with the Notes Reuse team.

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Three Persistency Approaches — Table Structures

Generic Basis table vs Application-specific own table vs Current ZPO_LONGTEXT

Approach 1 — Generic Table: SGBT_NTCEONT (Basis/BCSRV Team)

MANDT	CLNT 3	Client
OBJECT_TYPE	CHAR 10	Business Object Type — generic, not EKKO/EKPO specific
OBJECT_ID	CHAR 70	Generic document ID — CHAR70 (same limitation as STXH TDNAME)
NOTE_ID	CHAR 4	Note/Text type
LANGUAGE	LANG 1	Language key
NOTE_CONTENT	STRG —	Long string — plain text content
CREATED_BY	CHAR 12	Created by user
CREATED_AT	DATS 8	Created date
CHANGED_BY	CHAR 12	Last changed by
CHANGED_AT	DATS 8	Last changed date

Owner: BCSRV component (Basis team). Already used by Supplier Confirmation (new feature, no migration needed). Notes Reuse RAP Business Object is built on this table. CDS views and C1 contract views owned by Basis — not by PO team. CHAR70 OBJECT_ID = same generic limitation as STXH TDNAME.

Approach 2 — Application-Specific Own Table: SDTP_TEXT (Sales Reference)

MANDT	CLNT 3	Client
SD_DOCUMENT_ID	CHAR 10	Properly typed Sales Order number (= VBELN) — NOT generic CHAR70
SD_DOCUMENT_ITEM	NUMC 6	Item number (= POSNR) — properly typed
TEXT_OBJECT	CHAR 10	VBBK (header) or VBBP (item)
TEXT_ID	CHAR 4	Text type (0001, 0002 etc.)
LANGUAGE	LANG 1	Language key
TEXT_CONTENT	STRG —	Plain text string — converted from ITF format via CONVERT_ITF_TO_STREAM_TEXT
REF_TEXT_OBJECT	CHAR 10	Reference chain — source object
REF_TEXT_ID	CHAR 4	Reference chain — source text ID
REF_TEXT_NAME	CHAR 70	Reference chain — source text name
CREATED_BY/AT	CHAR+DATS	Administrative fields
CHANGED_BY/AT	CHAR+DATS	Administrative fields
AUTH_ENCODED	CHAR ~20	Auth fields encoded via TDTITLE: DocCategory+DocType+SalesOrg+DistCh+Division
SDM_VERSION	NUMC 2	Migration status — in root table VBAK (not in text table itself)

Owner: SD LOB team. Handler class registered in TTXOB for VBBK/VBBP. SAP Script X team explicitly recommended this approach for all LOBs. Sales completed PoC in CE2608. SDM migrated 2.67M records in CCW/720 in 3.5 hours. ITF format converted to plain text string during migration.

Approach 3 — Current Lift & Shift Table: ZPO_LONGTEXT		
MANDT	CLNT 3	Client
EBELN	CHAR 10	PO Number — properly typed ✓
EBELP	NUMC 5	Item number — 00000 = header, 00010 = item 10
TDOBJECT	CHAR 10	EKKO (header) or EKPO (item)
TDID	CHAR 4	F01, F02, F09 etc.
TDSRAS	LANG 1	Language key: D confirmed for this system
LINE_COUNTER	NUMC 5	Line sequence 00001, 00002... — ORDER BY field
TDFORMAT	CHAR 2	Format indicator: * = normal text line
TDLINE	CHAR 132	PLAIN TEXT content — one 132-char line per row (NOT a string)
MIGRATED_AT	DATS 8	Date row was written
MIGRATED_BY	CHAR 12	User who wrote this row (sy-uname)
SOURCE	CHAR 1	D = Dual-write by BAdI, M = Migration report

Integration: BAdI ME_PROCESS_PO_CUST fires AFTER SAVE_TEXT. No SDM_VERSION tracking, no handler class, no automatic read/write switching. Always dual-write (BAdI always fires). PoC/Lift & Shift approach only. EKPO TDNAME confirmed: NO space between EBELN and EBELP on this system (e.g. 450007774400010).

ITF Format Conversion (Critical for Migration)

SAPscript stores text in ITF (Interchange Text Format) — a proprietary SAP format. The new persistency stores PLAIN TEXT strings. Conversion happens during migration:

```
" ITF table (TLINE rows) → Plain text string

CALL FUNCTION 'CONVERT_ITF_TO_STREAM_TEXT'

  EXPORTING i_type = 'TEXT'  TABLES i_text = lt_tline_table

  IMPORTING e_text = lv_plain_string.

" Plain text string → ITF table (for reverse – e.g. display in editor)

" Step 1: Split string into 132-char chunks

" Step 2: Convert

CALL FUNCTION 'CONVERT_STREAM_TO_ITF_TEXT'

  EXPORTING i_type = 'TEXT'  TABLES i_text = lt_stream_table

  IMPORTING e_text = lt_tline_table.
```

Authorization Encoding via TDTITLE Field

Auth fields are application-specific — not part of the SAPscript framework interface. Confirmed by text framework colleague Rene Zink: THEAD-TDTITLE field can be reused to pass application-specific data to the handler class.

Auth Field	From	Example Value
SD Document Category	VBAK-VBTYP	C
Sales Document Type	VBAK-AUART	TA
Sales Organization	VBAK-VKORG	1000
Distribution Channel	VBAK-VTWEG	10
Division	VBAK-SPART	00

```
" Concatenated value in TDTITLE (Option 1 – preferred):

" SD Doc Category + Sales Doc Type + Sales Org + Dist Ch + Division

" Example: 'C   TA  10001000'  (values padded with spaces)

" PO equivalent would be:

" Purch Org + Company Code + Document Type + ... encoded similarly
```

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Full Comparison — All Three Approaches

Every architectural aspect side by side

Aspect	RAP Generic Table (SGBT_NTCEONT)	Sales SDD Own Table (SDTP_TEXT)	ZPO_LONGTEXT (Current)
Table owner	Basis / BCSRV	SD LOB team	Lift & Shift / PO team
Table name	SGBT_NTCEONT	SDTP_TEXT	ZPO_LONGTEXT
Key design	CHAR70 OBJECT_ID (generic)	Typed VBELN + POSNR ✓	Typed EBELN + EBELP ✓
Text storage	Long string (plain)	Long string (ITF→plain)	One row per 132-char line
Integration method	RAP Business Subject in BO model	Handler class in TTXOB	BAdI ME_PROCESS_PO_CUST
SAVE_TEXT routing	Not used — RAP APIs only	Auto via framework	Separate hook after SAVE_TEXT
Migration mechanism	N/A — new feature only	SDM CL_SDM_PACKAGE_MIGRATION	Manual report — no SDM
Dual-write control	N/A	Auto via IS_MIGRATION_FINISHED	Always on — BAdI always fires
Read switching	N/A	Auto at SDM completion	No cutover logic — always reads ZPO
CDS views	Built by Basis (C1 released)	Built by SD team on SDTP_TEXT	Not built yet
Analytics/BW	Via Basis CDS views	Via SD CDS views	Not supported
Auth fields	Not possible	YES — via TDTITLE encoding	Not stored
Text references	Not supported (confirmed)	Stored in REF_ fields	Not handled
App-specific attributes	NOT possible	YES — any field	YES — custom table
SDM_VERSION in root	Not required	Required in VBAK	Not required
Notes Reuse RAP compat.	YES — designed for it	UNKNOWN (open question)	UNKNOWN (open question)
SAP Script X recommendation	Not mentioned	YES — recommended	Aligned with this
Production ready	Yes — Supplier Confirmation	Yes — Sales CE2608 PoC	PoC only

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Recommendation for Standard PO (EKKO / EKPO)

Sales SDD own table approach — pending Notes Reuse compatibility confirmation

RECOMMENDED: Sales SDD approach — Application-specific own table + SAPScript Handler Class Framework + SDM (Silent Data Migration) PENDING: Confirm with Notes Reuse team that own table is compatible with Notes Reuse RAP BO

Why NOT Generic Table (SGBT_NTCEONT) for PO

Concern	Detail
No migration path for existing texts	SGBT_NTCEONT is for NEW business objects only. PO has millions of existing texts in STXH/STXL needing migration.
CDS views owned by Basis	PO team cannot add EBELN-specific joins or PO analytics without Basis involvement.
CHAR70 OBJECT_ID	Still generic — same JOIN difficulty as STXH TDNAME. No EBELN typed key.
No PO-specific fields	Cannot add Purchase Org, Document Type for authorization or reporting.
References not supported	Direct text references confirmed NOT supported in new persistency (SDD confirmed).

Why Sales SDD Own Table IS Recommended for PO

Benefit	PO-Specific Reasoning
Typed keys EBELN + EBELP	Direct SQL joins with EKKO, EKPO — no CHAR70 parsing needed
PO team owns the table	Can add EKKO-specific fields: Purchasing Org, Company Code, Doc Type for auth
SAP Script X recommended	Explicit guideline to all LOBs — PO follows same path as Sales
SDM framework proven	Sales completed this in CE2608 — reusable pattern for PO in CE2708
Automatic read/write switching	IS_MIGRATION_FINISHED handles cutover — no manual steps needed
AI and BDC ready	Plain text string in single field — AI can process directly
CDS views on PO terms	Analytics CDS can join with I_PurchaseOrderBasic, I_PurchaseOrderItem directly

Proposed PO Table: EKKT_TEXT

EKKT_TEXT — Recommended PO New Text Table		
MANDT	CLNT 3	Client
EBELN	CHAR 10	PO Number — properly typed (NOT generic CHAR70) ✓
EBELP	NUMC 5	Item — 00000 = header, 00010 = item 10 ✓
TEXT_OBJECT	CHAR 10	EKKO (header) or EKPO (item)
TEXT_ID	CHAR 4	F01, F02, F09 etc.
LANGUAGE	LANG 1	Language key: D for this system
TEXT_CONTENT	STRG —	Plain text string — ITF converted via CONVERT_ITF_TO_STREAM_TEXT
REF_TEXT_OBJ	CHAR 10	Reference chain — source object
REF_TEXT_ID	CHAR 4	Reference chain — source text ID
REF_TEXT_NAME	CHAR 70	Reference chain — source text name
CREATED_BY/AT	CHAR+DATS	Administrative — created by/date
CHANGED_BY/AT	CHAR+DATS	Administrative — changed by/date
AUTH_PORG	CHAR 4	Purchase Organization — explicit auth field (better than encoding)
AUTH_BUKRS	CHAR 4	Company Code — explicit auth field
AUTH_BSART	CHAR 4	Document Type — explicit auth field

KEY DIFFERENCE from ZPO_LONGTEXT: TEXT_CONTENT = full plain text STRING (not one row per 132 chars — no LINE_COUNTER needed) Auth fields stored EXPLICITLY (not encoded via TDTITLE like Sales) Reference fields stored No TDFORMAT / TDLINE / LINE_COUNTER columns

Migration Sequence

Release	Task
CE2702 — PoC	Clarify Notes Reuse compatibility. Design EKKT_TEXT. Implement ZCL_PO_RSTXT_PERSISTENCE handler class. Register in TTXOB for EKKO and EKPO. Add SDM_VERSION to EKKO. Implement ZCL_SDM_EKKO_TEXT_MIGRATION. POC: write to both STXH and EKKT_TEXT via modernized OData V4 API.
CE2708 — Full	SDM runs in background — migrates all historic PO texts. IS_MIGRATION_FINISHED → TRUE → read switches to EKKT_TEXT. Build CDS views. Plug notes into PO OData V4 API. Regression test for all EKKO/EKPO business objects.

ANALOGY: Think of the Handler Class as a TRAFFIC CONTROLLER standing at the SAVE_TEXT/READ_TEXT junction. It redirects traffic to the new table. Without it, all traffic goes to STXH/STXL as before.

What It Is

A runtime plug-in registered in table TTXOB that intercepts every SAVE_TEXT and READ_TEXT call and routes them to your new table instead of STXH/STXL. It is a class that inherits from CL_RSTXT_PERSISTENCE_FRAMEWORK.

WITHOUT Handler Class (before registration):

SAVE_TEXT → always writes to STXH/STXL

READ_TEXT → always reads from STXH/STXL

WITH Handler Class registered in TTXOB for EKKO and EKPO:

SAVE_TEXT → SAP framework checks TTXOB → calls ZCL_PO_RSTXT_PERSISTENCE

→ your CREATE or CHANGE method runs

→ writes to EKKT_TEXT

→ AND writes to STXH/STXL if IS_MIGRATION_FINISHED = FALSE

READ_TEXT → SAP framework calls ZCL_PO_RSTXT_PERSISTENCE→READ

→ if IS_MIGRATION_FINISHED = FALSE → reads STXH/STXL

→ if IS_MIGRATION_FINISHED = TRUE → reads EKKT_TEXT

5 Methods to Implement

Method	When Called	What It Does
CREATE	Every SAVE_TEXT creating new text	Write new text to EKKT_TEXT (+ STXH during migration)
CHANGE	Every SAVE_TEXT updating text	Update text in EKKT_TEXT (+ STXH during migration)
DELETE	Every DELETE_TEXT call	Remove text from EKKT_TEXT (+ STXH during migration)
READ	Every READ_TEXT call	Read from EKKT_TEXT (new) or STXH/STXL (old) based on IS_MIGRATION_FINISHED
READ_HEADERS_VIA_RANGES	Every SELECT_TEXT call	Read text headers from EKKT_TEXT or STXH

Registration in TTXOB

Table TTXOB – Text Object definition:

TDOBJECT = 'EKKO' → HANDLER_CLASS = 'ZCL_PO_RSTXT_PERSISTENCE'

TDOBJECT = 'EKPO' → HANDLER_CLASS = 'ZCL_PO_RSTXT_PERSISTENCE'

After registration – ALL applications using SAVE_TEXT/READ_TEXT for EKKO/EKPO

automatically route through your handler class – no code changes in ME21N/ME22N needed.

Alternative: Register at TDID level in TTXID (per text type) instead of all TDIDs.

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SDM — Silent Data Migration

Background batch job that moves EXISTING historic texts from STXH/STXL to new table

ANALOGY: Think of SDM as REMOVAL MEN moving all the furniture from your old house to your new house while you are still living there. The Handler Class is the new house address label (new deliveries go to new house). But someone still has to move everything already in the old house. That is what SDM does.

What Problem SDM Solves

After Handler Class is registered:

NEW texts (created after registration):

→ Handler Class routes them to EKKT_TEXT immediately ✓

EXISTING historic texts (created BEFORE registration):

→ Still ONLY in STXH/STXL ✗

→ PO 4500001234 from 2020 — its text is ONLY in STXH/STXL

SDM migrates all the HISTORIC data in background — silently.

SDM Step by Step — What It Actually Does

Step	Action	Detail
1	Select package	SELECT ebeln FROM ekko WHERE sdm_version < target UP TO 30,000 ROWS
2	Read old texts	CALL FUNCTION 'SELECT_TEXT' → get text headers CALL FUNCTION 'READ_TEXT_TABLE' → mass read text content (decompresses STXL)
3	Convert format	CALL FUNCTION 'CONVERT_ITF_TO_STREAM_TEXT' → ITF table → plain text string
4	Delete existing	DELETE FROM ekkt_text WHERE ebeln IN @lt_package — ensures restartability
5	Insert new	INSERT ekkt_text FROM TABLE lt_new_texts — plain text rows in new table
6	Update status	UPDATE ekko SET sdm_version = target WHERE ebeln IN @lt_package
7	Repeat	Process next package until all POs have sdm_version = target

Performance Numbers from Sales (Reference)

System	VBAK Records	Text Records	Package Size	Duration
DDCI (test)	40,853	340	30,000	Fast
CCW/720 (perf)	13.9 million	2.67 million	30,000	3.5 hours

Open SQL limit: max 32,767 entries in WHERE range table → package size set to 32,000. Internal packaging within SELECT_TEXT calls every 30,000 documents. Old STXH/STXL records are NOT deleted during SDM — kept as safety net for restart. PO estimate: if similar volume to Sales (~13M POs) → ~3.5 hours SDM duration.

IS_MIGRATION_FINISHED — The Bridge Between Handler and SDM

IS_MIGRATION_FINISHED = FALSE (SDM still running — called in handler class):

Handler CREATE/CHANGE → writes to STXH/STXL AND EKKT_TEXT (dual-write)

Handler READ → reads from STXH/STXL (complete data – SDM still running)

IS_MIGRATION_FINISHED = TRUE (SDM complete – all data in new table):

Handler CREATE/CHANGE → writes to EKKT_TEXT ONLY

Handler READ → reads from EKKT_TEXT (new – complete)

Implementation:

```
METHOD is_migration_finished.
```

```
    rv_finished = cl_sdm_proc_status_api=>is_sdm_finished(
```

```
        i_sdm_name = 'ZCL_SDM_EKKO_TEXT_MIGRATION' ).
```

```
ENDMETHOD.
```

Result buffered for entire dialog session – avoids repeated SELECT statements.

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Handler Class vs SDM — Key Differences

They are two completely different mechanisms that work together

They Are NOT the Same Thing: Handler Class = RUNTIME routing (where does NEW text go?) SDM = BACKGROUND migration (how do EXISTING historic texts get moved?)

Aspect	SAPScript Handler Class Framework	SDM — Silent Data Migration
What it is	Runtime plug-in registered in TTXOB	Background batch migration job
What it solves	Routes NEW SAVE_TEXT/READ_TEXT to new table	Moves EXISTING historic data to new table
When it runs	Every time SAVE_TEXT or READ_TEXT is called	Once in background — packages of 30,000
How activated	Register handler class in TTXOB.HANDLER_CLASS	Execute SDM job from SDM cockpit
Duration	Instant — runs per transaction	Hours (3.5h for 13M records)
Who drives it	SAP Script framework — automatic	SDM cockpit — background job
Controls read switch	YES — via IS_MIGRATION_FINISHED	NO — it just migrates data
SAP class to inherit	CL_RSTXT_PERSISTENCE_FRAMEWORK	CL_SDM_PACKAGE_MIGRATION
Sales class name	Part of SD text persistency class	CL_SDM_SD_VBAK_TEXT_MIGRATION
PO equivalent to build	ZCL_PO_RSTXT_PERSISTENCE	ZCL_SDM_EKKO_TEXT_MIGRATION

Full Timeline — How They Work Together

BEFORE any change:

STXH/STXL has all PO texts (old house – all furniture here)

EKKT_TEXT is empty (new house – empty)

STEP 1 – Register Handler Class in TTXOB for EKKO/EKPO:

New texts → go to EKKT_TEXT (AND STXH/STXL while SDM running)

Historic texts → still only in STXH/STXL

IS_MIGRATION_FINISHED = FALSE → reads still from STXH/STXL

STEP 2 – SDM runs in background (hours/days):

Package by package: reads STXH/STXL → converts → inserts EKKT_TEXT

Updates SDM_VERSION in EKKO for each completed package

System continues working normally throughout

STEP 3 – SDM completes (all SDM_VERSION updated):

IS_MIGRATION_FINISHED = TRUE

Handler Class now reads from EKKT_TEXT

```
Handler Class now writes to EKKT_TEXT only  
  
STXH/STXL = frozen/legacy (kept as safety net)
```

STEP 4 – After validation period:

```
STXH/STXL rows for PO texts cleaned up (archiving)  
  
EKKT_TEXT is the single source of truth ✓
```

In One Sentence Each

Mechanism	One-Line Definition
SAPScript Handler Class Framework	A plug-in that intercepts every SAVE_TEXT and READ_TEXT call and routes it to your new table — effective immediately for all new transactions.
SDM (Silent Data Migration)	A background job that moves all the existing historic text data from STXH/STXL to your new table silently while the system keeps running normally.